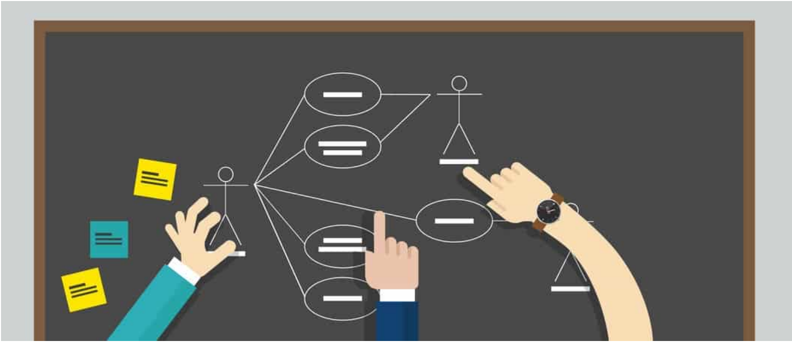
Chapter 6:
Use Case Diagram - 1



From user requirement document to use story

User stories are product specifications told on behalf of a user.

It can be written as:

As a < type of user >, I want < some goal > so that < some reason >.

As a **driver**, I want to block badly behaved passengers so they are never shown me again.

As a **passenger**, I want to link the credit card to my profile so that I can pay for a ride faster, easier and withot cash.

As a **driver**, I want to add photos of my car in my profile so that I can attract more users.

As a **passenger**, I want several available drivers to be displayed so that I can choose the most suitable option for me.

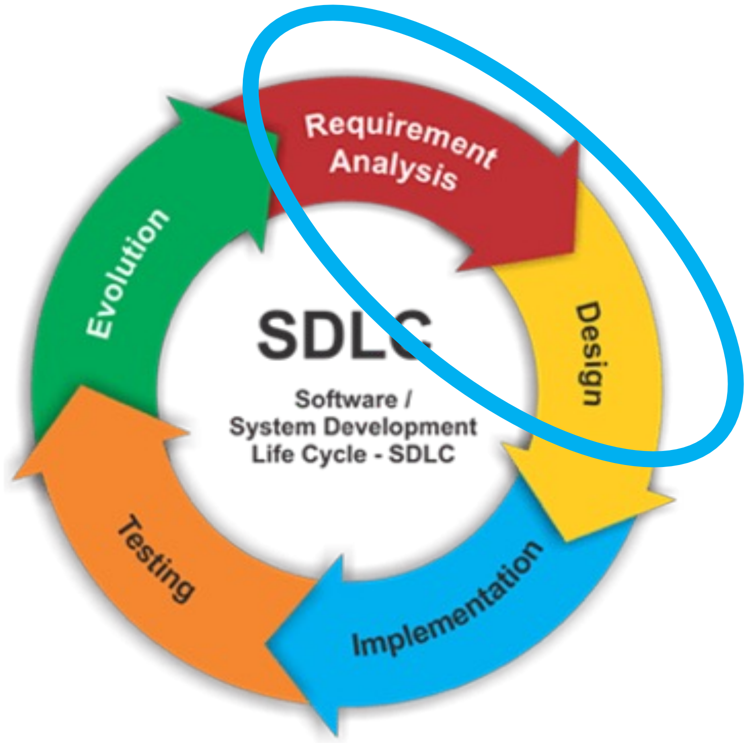
From user story to use case

Use Case is a written description of the user's interaction with the software product to accomplish a goal.

- Use case example from a Food Delivery App:
  - As a user, I can order food from Shops.
  - As a user, I can change my contact details.
  - As a power user, I can cancel orders.
  - As a power user, I can reassign riders to deliver food.

Use case in SDLC

Use cases are realised in analysis and design phases of the SDLC.



UML

UML stands for Unified Modeling Language.

Commonly used by software developers/programmers.

UML is used to design diagrams to visualise a project.

Different types of UML diagrams

Use case diagram

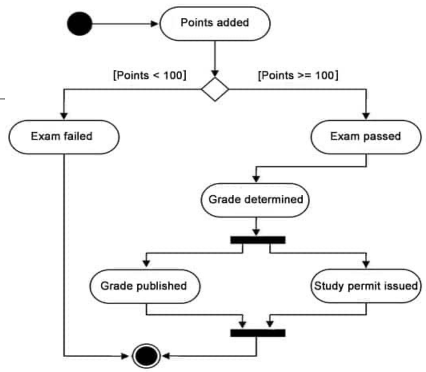
Class diagram

State diagram

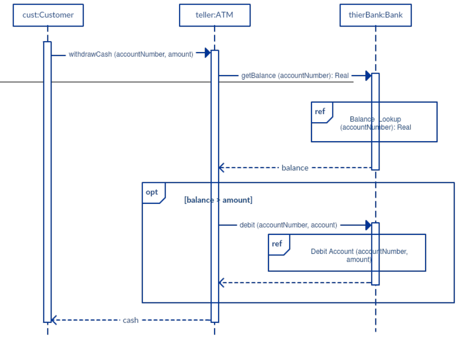
Object diagram

Sequence diagram

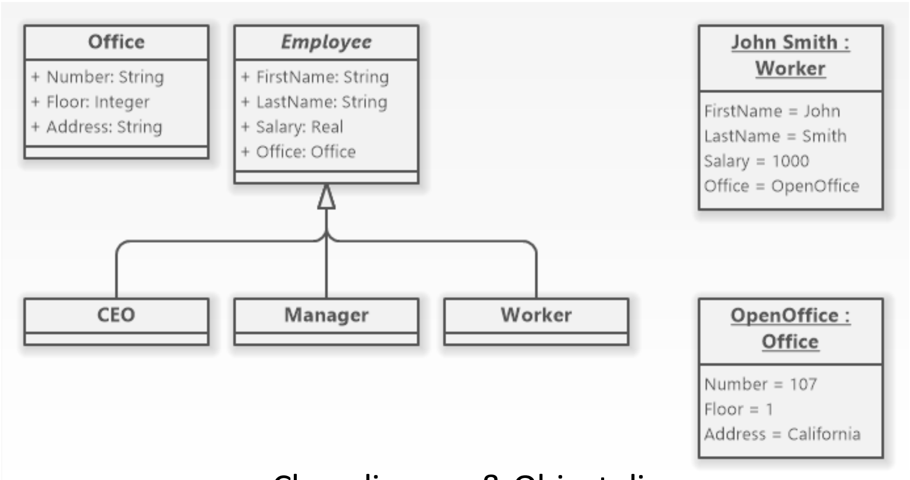
...



State diagram



Sequence diagram



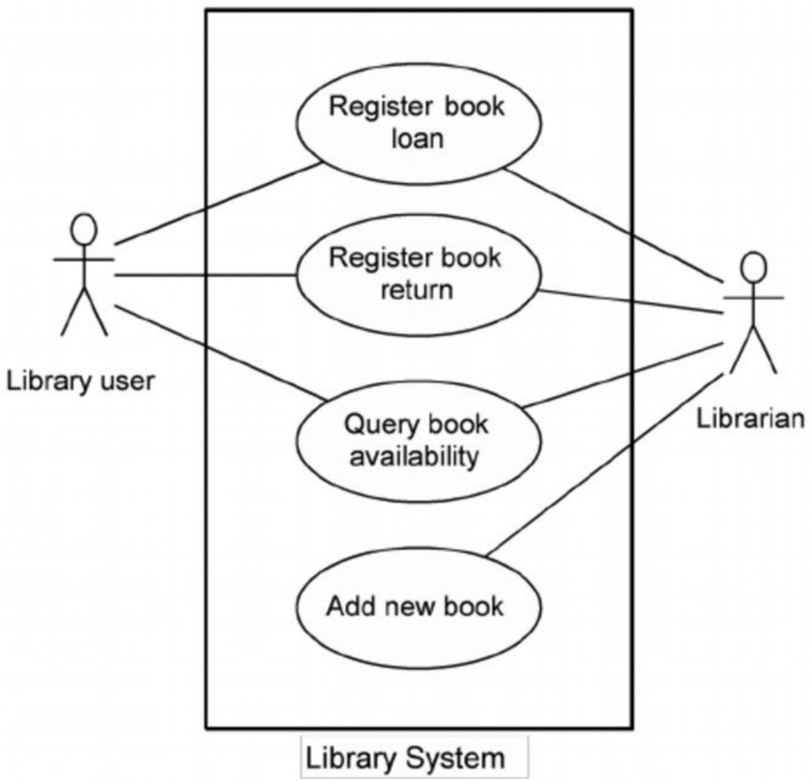
Class diagram & Object diagrams

A picture is worth a thousand words.

Combining use cases into a use case diagram

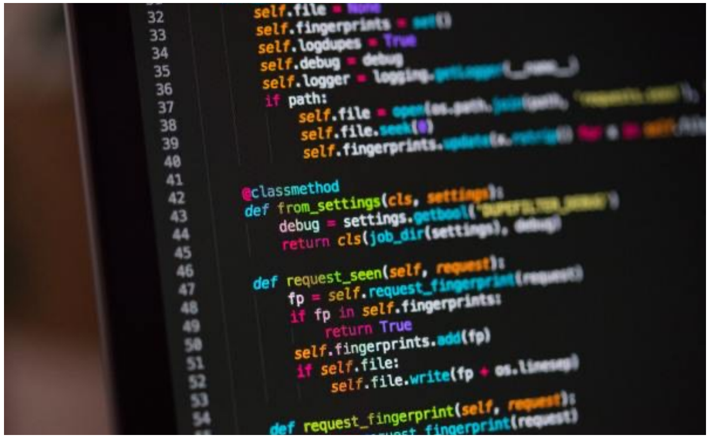
The overall list of the system's use cases can be drawn as a high-level diagram known as use case diagram.

Each oval in the diagram represents a single use case.



Why use case diagram?

- A great way to communicate complex ideas with other team members.
    - Especially with a <u>non-technical person</u> such as a business representative.



VS

What is a use case diagram?

A graphical notation that help in designing and communicating software systems.

Does not describe how the functionalities are implemented.

Illustrates the <u>different types of roles in a system</u> and how those roles <u>interact</u> with the system.

Use case diagram elements and notations

System boundary

- Sets a system’s scope to use cases.

Actor

- User that interacts with a system.

Use case

- Written description of the user's interaction with the software product to accomplish a goal.

Relationships

- Association is the relationship between an actor and a use case.
- More complicated relationships will be discussed in the next chapter

System boundary

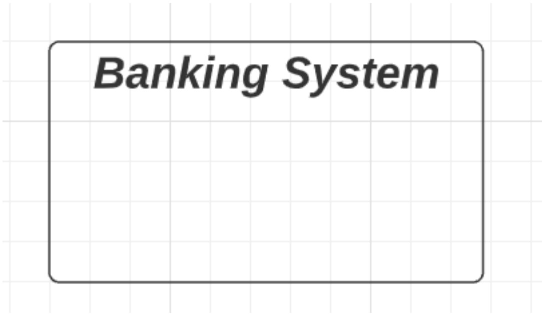
A box with name.

System boundary sets a <u>system's scope</u> to <u>use cases.</u>

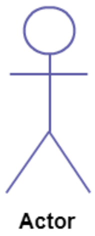
All use cases outside the box would be considered outside the scope of that system.

System can be

- Website
- Software
- APP
- Etc.



Actor



Stick figures with name.

The <u>users</u> that <u>interact with a system</u>.

An actor can be a person, an organisation, or an outside system that interacts with your application or system.

Named by noun.

Give meaningful names for actors

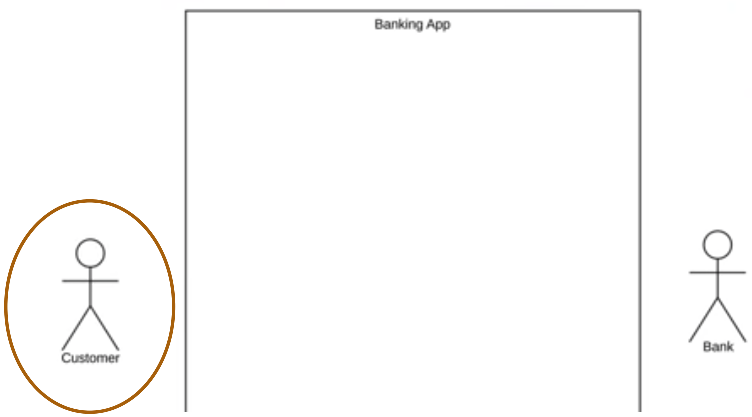
- Name it with the function rather than the organisation name. (Eg: Airline Company is better than PanAir)

Primary Actors (Active Actors)

Primary Actors (also called active actors)

- Initiate the use of the system.
- Placed on the left side of the system so that they can be quickly highlighted as important roles in the system.

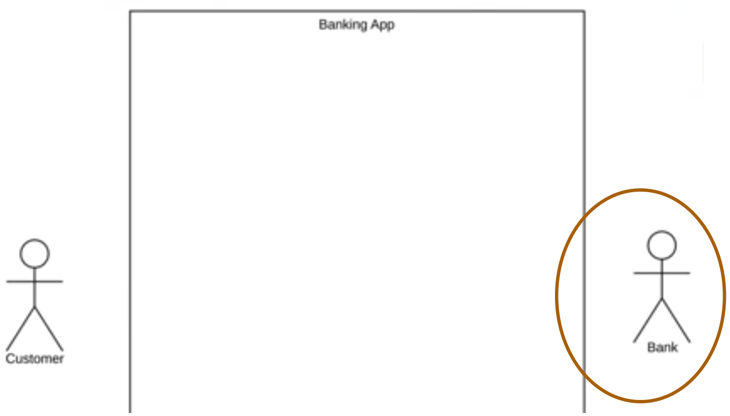
If there are a few primary actors, it is okay to place primary actors on <u>any sides</u> of the system.



Secondary Actors (Passive Actors)

Secondary Actors (also called passive actors)

- React to the use case.
- Do not interact with the system on their own.
- Provide a service to the system.
- Placed on the right side of the system.
- Example : database, bank



Identify Primary and Secondary Actors

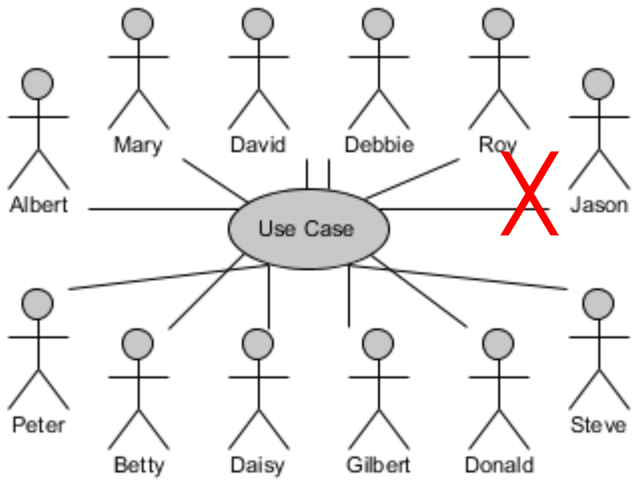
1. A user clicks the search button on an application's user interface.
2. The application sends and SQL query to a database system.
3. The database system responds with a result set.
4. The application formats and displays the result set to the user.

Primary actor
- The user
- He initiates the interaction with the system (application).

Secondary actor
- The database system
- The application initiates the interaction by sending an SQL query to the database

Actor is a role, not a real person

- Actors don't have names. They're not "Bob."
- They represent the role of someone interacting with the system
- Must be external to the system



- Website User
- Webmaster
- Writer
- Editor
- Publisher
- Bookseller
- Traveller
- Student
- Registrar
- Email Server

Use case

Use case is a written description of the user's interaction with the software product to accomplish a goal.

- As a user, I can order food from Shops.
- As a power user, I can cancel orders.

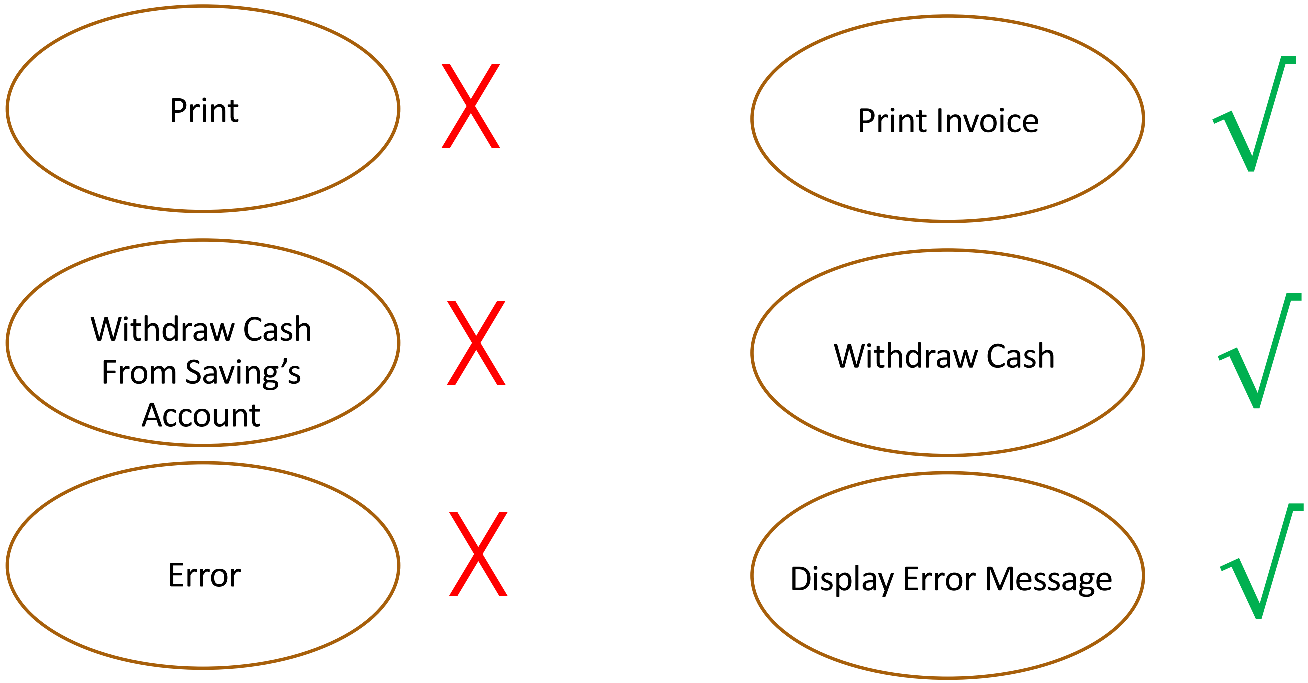
In UML, use case is represented by horizontally shaped oval or ellipse with name in it.

In UML, use case is named by verb.

When thinking of use cases, think of the end goal of a user.

Try to list the use cases in a logical order.

Use cases with short and meaningful names

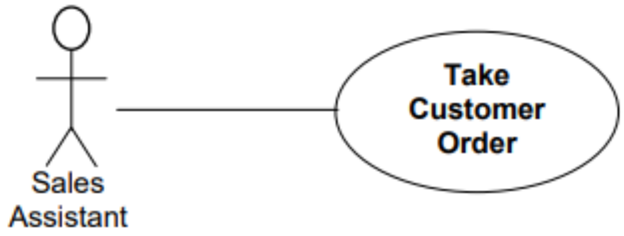


Relationship

An association is the <u>relationship</u> between <u>an actor and a use case</u>.

It indicates that an actor can use a certain functionality of the system.

Association is represented by a solid line between actor and use case



The following more complicated relationships will be discussed in the next chapter

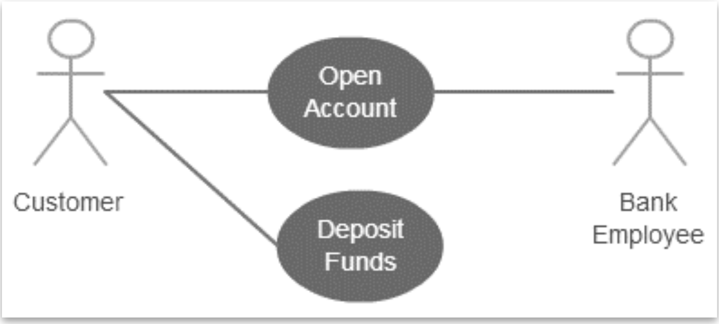
- Extend between two use cases
- Include between two use cases
- Generalisation of an actor
- Generalisation of a use case

Association between Use Cases and Actors

An actor must be associated with at least one use case.

An actor can be associated with multiple use cases.

Multiple actors can be associated with a single use case.



How to create use case diagram

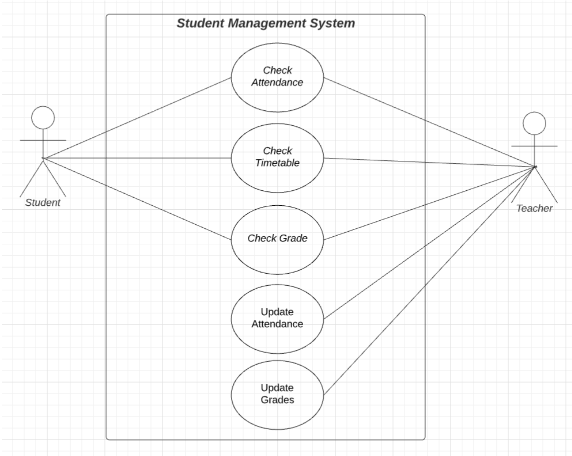
1. Set the scope of the system
2. Identify the Actors (role of users) of the system.
3. Identify what are the users required the system to perform to achieve these goals.
4. Create use cases for every goal.
5. Structure the use cases. (to be discussed further in next chapter)
6. Prioritise, review, validate and refine the use cases.

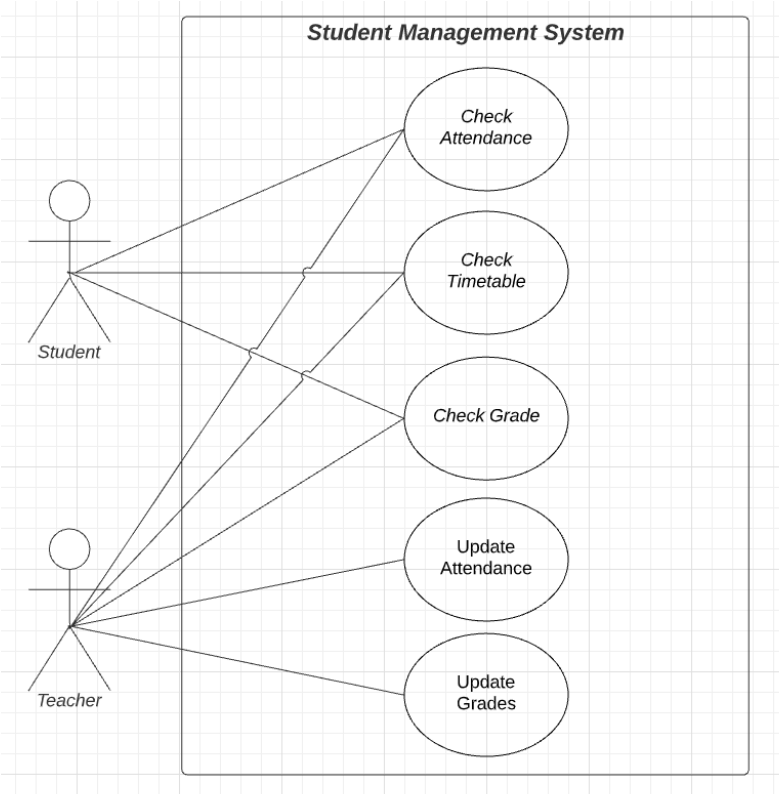
Exercise: Student Management System

In a Student Management System, identify the main users.

List done the use cases.

Draw the use case diagram.





Summary

UML

- Unified Modeling Language
- Used by Software programmers to design diagrams to visualise a project

Use case diagram

- Requirement Analysis and Design phases of SDLC
- Commonly used by software developers/programmers
- Can be used for communicating idea with a non-technical person
- Illustrates the different types of roles in a system and how those roles interact with the system.

Summary

Key elements in a use case diagram

1. System boundary
   - Sets a system’s scope to use cases.
2. Actor
   - User that interacts with a system.
   - Primary actor initiates the use of the system.
   - Secondary actor reacts to the use case.
   - Primary actor is usually placed on the left side of the system.
   - Secondary actor is placed on the right side of the system.
   - If there are a few primary actors, it is okay to place primary actors on any sides of the system
3. Use case
   - Written description of the user's interaction with the software product to accomplish a goal.
4. Relationship
   - Association is a relationship between an actor and a use case.

References:

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